

Hoisting

- Hoisting :- To move all the declaration of functions or variable to the top of the current scope, before execution of the code.
- We can access variable before their initialisation and functions before their declaration.
- No matter where variable and functions are declared, they move to the top of their scope, regardless of whether their scope is global or local.
- Hoisting is a default behaviour of JavaScript.
- Declaration only hoist, not an initialisation.
- Let and Const are also hoisted, but we can't access them until they are assigned, because they are in temporal Dead Zone (TDZ). To avoid TDZ, need to declare let & const to the top of a program. JS allocate memory to all the variable & function before execution.
When const get hoisted, it will define (or assign a value) 'undefined', at memory creation phase, but during code execution, it will assign a 'value'. This violates the meaning of const. This is the reason, why CONST can't hoist. and LET got tagged along with it.
- Variable can use before declaration.
- Function can call before function definition.
So, can access variable & function before initialisation without any error.


```

console.log(x); // undefined // 1
greeting(); // { console.log("Hi greeting"); // Hi greeting
greeting1(); // uncaught TypeError: greeting1 is not a function
greeting2(); // because it's treating like a variable not a function

```

So, rather than storing a function body, it's storing undefined by default.

```

console.log(greeting); // { console.log("Hi greeting");
console.log var x=1;
function greeting() {
  console.log("Hi greeting");
}
function var greeting1 = function () {
  console.log("Hey greeting1");
}
var greeting2 = () => {
  console.log("Hello greeting2");
}

```

→ Normal function
→ Function Expression
→ Arrow function

```

console.log(x) // 1
greeting() // Hi greeting
greeting1() // Hey greeting1
greeting2() // Hello greeting2

```

(Temporal Dead Zone) (Undefined) (Hoisted)
(i.e Reference Error) ↑ give ↑ give

Before Declaration	let / const	var
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examples: - (Declaration hoist, initialization (ant))

① `var x`
`console.log(x) // undefined`
`x = 7`

② `var x = 7;`
`console.log(x) // 7`
Dt: / /

③ `x = 7;`
`console.log(x) // 7`
`var x;`

④ `console.log(x) // undefined`
`x = 7;`
`var x;`

(Uncaught Reference Error;

⑤ `console.log(x) // undefined`
`var x = 7;`

⑧ `console.log(x); // x is not defined`

(Missing initializer
in (const declaration)

⑥ `let x = 7;`
`console.log(x) // 7`

⑦ `const x;`
`console.log(x) // Error`
`x = 7;`